Smart Home AIoT – Test Data Collection Sheet – Phase B – Sensor Reading

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Generated: 2026-05-25 22:46

Instructions for use:
1. Run each test 10 times consecutively (or across sessions, as practical)
2. Record: Date, Time, exact Input given, Actual Output observed, Response Time (use stopwatch), and Pass/Fail
3. Pass criteria: Actual matches Expected description
4. After 10 trials, calculate: Pass Rate %, Average Response Time, Standard Deviation
5. Use Comments for any anomalies or special observations

Statistical formulas:
- Pass Rate = (Pass Count / 10) × 100%
- Avg RT = (response times) / 10
- Variance = $(xi -)^2$ / n
- Std Dev = √variance

Coverage Matrix

Phase	Description	Count
Phase B	Phase B – Sensor Reading	18
	TOTAL	18

Phase B — Sensor Reading

B01	Bedroom Temperature + Humidity (DHT22)	Phase B

Method: ดู HA Dashboard sensor.bedroom_temperature, humidity

Expected: 20-35°C, 40-80% (sanity range)

#	Date	Time	Input/Command	Actual Output	RT (ms)	P/F	Notes
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							

Pass Count: ___/10	Pass Rate: ____%	Avg RT: ____ ms	Std Dev: ____ ms

Comments:

B02 Living Room Temperature + Humidity Phase B

Method: ดู HA sensor.living_room_temperature, humidity

Expected: same range

#	Date	Time	Input/Command	Actual Output	RT (ms)	P/F	Notes
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							

Pass Count: ___/10	Pass Rate: ____%	Avg RT: ____ ms	Std Dev: ____ ms

Comments:

B03 Kitchen Temperature + Humidity Phase B

Method: ดู HA sensor.kitchen_temperature, humidity

Expected: same range

#	Date	Time	Input/Command	Actual Output	RT (ms)	P/F	Notes
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							

Pass Count: ___/10	Pass Rate: ____%	Avg RT: ____ ms	Std Dev: ____ ms

Comments:

B04	Bedroom LDR	Phase B

Method: ดู sensor.bedroom_ldr + เอามือบังเซ็นเซอร์ ค่าเพิ่ม

Expected: 0-4095 range (high=มืด)

#	Date	Time	Input/Command	Actual Output	RT (ms)	P/F	Notes
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							

Pass Count: ___/10	Pass Rate: ____%	Avg RT: ____ ms	Std Dev: ____ ms

Comments:

B05 Living Room LDR Phase B

Method: sanity test

Expected: 0-4095

#	Date	Time	Input/Command	Actual Output	RT (ms)	P/F	Notes
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							

Pass Count: ___/10	Pass Rate: ____%	Avg RT: ____ ms	Std Dev: ____ ms

Comments:

B06 Kitchen LDR Phase B

Method: sanity test

Expected: 0-4095

#	Date	Time	Input/Command	Actual Output	RT (ms)	P/F	Notes
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							

Pass Count: ___/10	Pass Rate: ____%	Avg RT: ____ ms	Std Dev: ____ ms
Comments:			

B07	Bathroom LDR	Phase B

Method: sanity test

Expected: 0-4095

#	Date	Time	Input/Command	Actual Output	RT (ms)	P/F	Notes
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							

Pass Count: ___/10	Pass Rate: ____%	Avg RT: ____ ms	Std Dev: ____ ms
Comments:			

B08 Garden LDR Phase B

Method: sanity test

Expected: 0-4095

#	Date	Time	Input/Command	Actual Output	RT (ms)	P/F	Notes
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							

Pass Count: ___/10	Pass Rate: ____%	Avg RT: ____ ms	Std Dev: ____ ms

Comments:

B09	Garage LDR	Phase B

Method: sanity test

Expected: 0-4095

#	Date	Time	Input/Command	Actual Output	RT (ms)	P/F	Notes
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							

Pass Count: ___/10	Pass Rate: ____%	Avg RT: ____ ms	Std Dev: ____ ms

Comments:

B10	Kitchen Gas (MQ-2)	Phase B

Method: จุดไฟแช็คใกล้ sensor.kitchen_gas_lpg ค่าขึ้น

Expected: ADC ขึ้นจาก baseline

#	Date	Time	Input/Command	Actual Output	RT (ms)	P/F	Notes
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							

Pass Count: ___/10	Pass Rate: ____%	Avg RT: ____ ms	Std Dev: ____ ms

Comments:

B11	Garage Smoke (MQ-2)	Phase B

Method: จุดไฟแช็ค/สเปรย์ sensor.garage_smoke_lpg ค่าขึ้น

Expected: ADC ขึ้นจาก baseline

#	Date	Time	Input/Command	Actual Output	RT (ms)	P/F	Notes
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							

Pass Count: ___/10	Pass Rate: ____%	Avg RT: ____ ms	Std Dev: ____ ms
Comments:			

Method: แตะดินเปียก vs แห้ง % เปลี่ยน

Expected: 0-100% range

#	Date	Time	Input/Command	Actual Output	RT (ms)	P/F	Notes
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							

Pass Count: ___/10	Pass Rate: ____%	Avg RT: ____ ms	Std Dev: ____ ms

Comments:

B13	Bathroom Water Leak (SNZB-05P)	Phase B

Method: แตะน้ำที่ขาทองแดง binary_sensor.bathroom_water_leak ON

Expected: binary state ON

#	Date	Time	Input/Command	Actual Output	RT (ms)	P/F	Notes
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							

Pass Count: ___/10	Pass Rate: ____%	Avg RT: ____ ms	Std Dev: ____ ms

Comments:

B14	Front Door Contact (SNZB-04)	Phase B

Method: เปิด/ปิดประตู binary_sensor.front_door ON/OFF

Expected: binary state เปลี่ยน

#	Date	Time	Input/Command	Actual Output	RT (ms)	P/F	Notes
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							

Pass Count: ___/10	Pass Rate: ____%	Avg RT: ____ ms	Std Dev: ____ ms

Comments:

B15	Bedroom Motion (PIR)	Phase B

Method: เดินผ่าน binary_sensor.bedroom_motion ON

Expected: ON 30s window

#	Date	Time	Input/Command	Actual Output	RT (ms)	P/F	Notes
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							

Pass Count: ___/10	Pass Rate: ____%	Avg RT: ____ ms	Std Dev: ____ ms

Comments:

B16 Living Room Motion (Zigbee PIR) Phase B

Method: เดินผ่าน binary_sensor.living_room_motion ON

Expected: ON

#	Date	Time	Input/Command	Actual Output	RT (ms)	P/F	Notes
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							

Pass Count: ___/10	Pass Rate: ____%	Avg RT: ____ ms	Std Dev: ____ ms

Comments:

B17	Bathroom Presence (LD2410)	Phase B

Method: เดินเข้า binary_sensor.bathroom_presence ON

Expected: ON

#	Date	Time	Input/Command	Actual Output	RT (ms)	P/F	Notes
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							

Pass Count: ___/10	Pass Rate: ____%	Avg RT: ____ ms	Std Dev: ____ ms

Comments:

B18	Garage Motion (PIR)	Phase B

Method: เดินผ่าน binary_sensor.garage_motion ON

Expected: ON 30s timeout

#	Date	Time	Input/Command	Actual Output	RT (ms)	P/F	Notes
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							

Pass Count: ___/10	Pass Rate: ____%	Avg RT: ____ ms	Std Dev: ____ ms

Comments: